MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, ANALOG, KEY CODE NGP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.25986 failures per million operating hours for class I or 0.90942 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NGP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/324-1 for class I and M28787/324-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

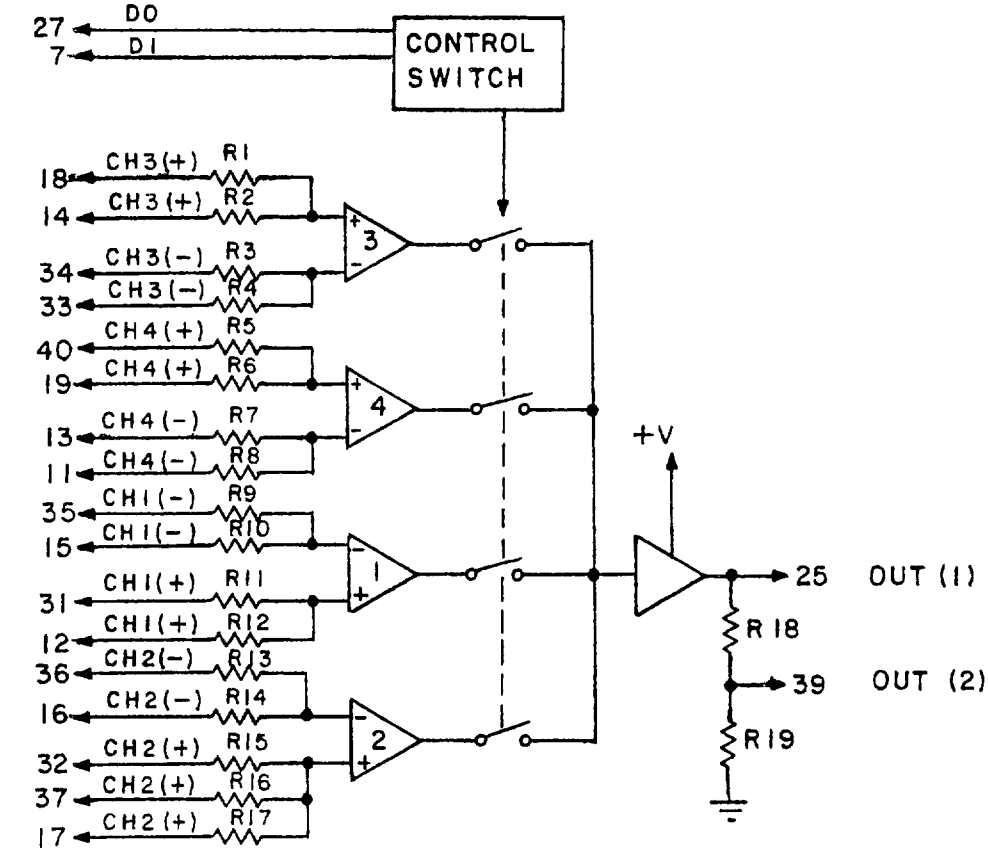
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

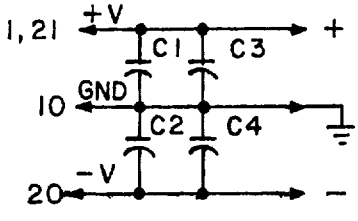
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. R1-R8 = R13-R16 = 51.1 kΩ ±0.5%, 1/10 W.
2. R9-R12 = 51.1 kΩ ±0.1%, 1/10 W.
3. R17 = 24.4 kΩ ±0.5%, 1/10 W.
4. R18 = 6.04 kΩ ±0.5%, 1/10 W.
5. R19 = 3.01 kΩ ±0.5%, 1/10 W.
6. C1 = C2 = 1 µF ±10%, 50 V.
7. C3 = C4 = 0.01 µF ±10%, 50 V.
8. Pins 2, 3, 4, 5, 6, 8, 9, 22, 23, 24, 26, 28, 29, 30, and 38 are unused.

FIGURE 1. NGP module function diagram, multiplexer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	V_S stress	III	Q	Must pass remaining tests										
B1	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B2-B16	Isolation													
	Initial	IV	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C	Output short circuit current	V	S	5	40	5	40	5	40	5	40	5	40	mA
D1,D2	Power supply current	VI	S	1	8	1	8	1	8	1	8	1	8	mA
E1,E2	Power supply transient amplitude	VII	Q	0	350	0	350	0	350	0	350	0	350	mV
F1,F2	Power supply filter capacitance	VIII	X	0.75	1.25	0.75	1.25	0.75	1.25	0.7	1.3	0.7	1.3	μF
G1-G4	Power supply rejection ratio check	IX	S	-∞	-52	-∞	-52	-∞	-52	-∞	-52	-∞	-52	dB
H1-H6	Resistance	X	X	101.4	103.0	101.4	103.0	101.4	103.0	101.4	103.0	101.4	103.0	kΩ
H7,H8	Resistance	X	X	102.0	102.4	102.0	102.4	102.0	102.4	102.0	102.4	102.0	102.4	kΩ
H9	Resistance	X	X	75.3	76.7	75.3	76.7	75.3	76.7	75.3	76.7	75.3	76.7	kΩ
H10	Resistance	X	X	9.0	9.1	9.0	9.1	9.0	9.1	9.0	9.1	9.0	9.1	kΩ
H11	Resistance	X	X	0	1	0	1	0	1	0	1	0	1	Ω
H12	Resistance	X	X	2.99	3.03	2.99	3.03	2.99	3.03	2.99	3.03	2.99	3.03	kΩ
J1-J4	Offset voltage	XI	X	-18	18	-18	18	-18	18	-18	18	-18	18	mV
K1,K2	Digital input current, high level	XII	X	0	15	0	15	0	15	0	15	0	15	nA
L1,L2	Digital input current, low level	XII	X	0.1	1.6	0.1	1.6	0.1	1.6	0.1	1.6	0.1	1.6	mA
M1-M5	Gain (200 kHz) check	XIII	S	-3	3	-3	3	-3	3	-3	3	-3	3	dB
N1-N8	Output slew rate check	XIV	X	2.6	8.0	2.6	8.0	2.6	8.0	2.6	8.0	2.6	8.0	V/μs

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P4	Distortion	XV	X	0	1	0	1	0	1	0	1	0	1	%
R1-R4	Common mode rejection ratio check	XVI	X	-∞	-50	-∞	-50	-∞	-50	-∞	-50	-∞	-50	dB
S1-S4	Crosstalk check	XVII	Q	-∞	-50	-∞	-50	-∞	-50	-∞	-50	-∞	-50	dB
T1-T4	Functional	XVIII	X	2.35	2.45	2.35	2.45	2.35	2.45	2.35	2.45	2.35	2.45	V

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test condition K shall be performed at 25°C, 60°C and 85°C using the initial limits for the applicable temperature.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B and D through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	300	300	mW
Supply voltages (see 6.5.1)	±20	±20	V
Output voltage swing	±10	±10	V

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	Class I	Class II	Unit
Input levels:			
Digital			
Low level	500 max	500 max	mV
High level	2.4 min	2.4 min	V
Analog	±20	±20	V
Minimum bandwidth	200	200	kHz

6.5 Module function. This module contains four operational amplifiers each containing the necessary input resistors for configuring unity gain differential amplifiers. Any one of these amplifiers can be selected through digital logic.

6.5.1 Maximum ratings. The maximum ratings for supply voltages given on 6.4 represent extremes beyond which damage to the module may occur. Operation at the extremes is performed in this specification to ensure that the module will survive these levels without damage.

6.5.2 Module handling procedures. This module contains devices which are susceptible to damage when exposed to a static electrical charge. When handling, shipping, or storing, all module pins should be shorted or the module should be placed in an electrically conductive package. Soldering iron tips and all test equipment must be grounded. Failure to comply with these guidelines could cause irreversible module damage.

6.5.3 Input to output relationship. The digital input versus ON channel relationship is shown in table II.

TABLE II. Input to output relationship. 1/ 2/

D1 (Pin 7)	D0 (Pin 27)	Selected channel
L	L	1
L	H	2
H	L	3
H	H	4

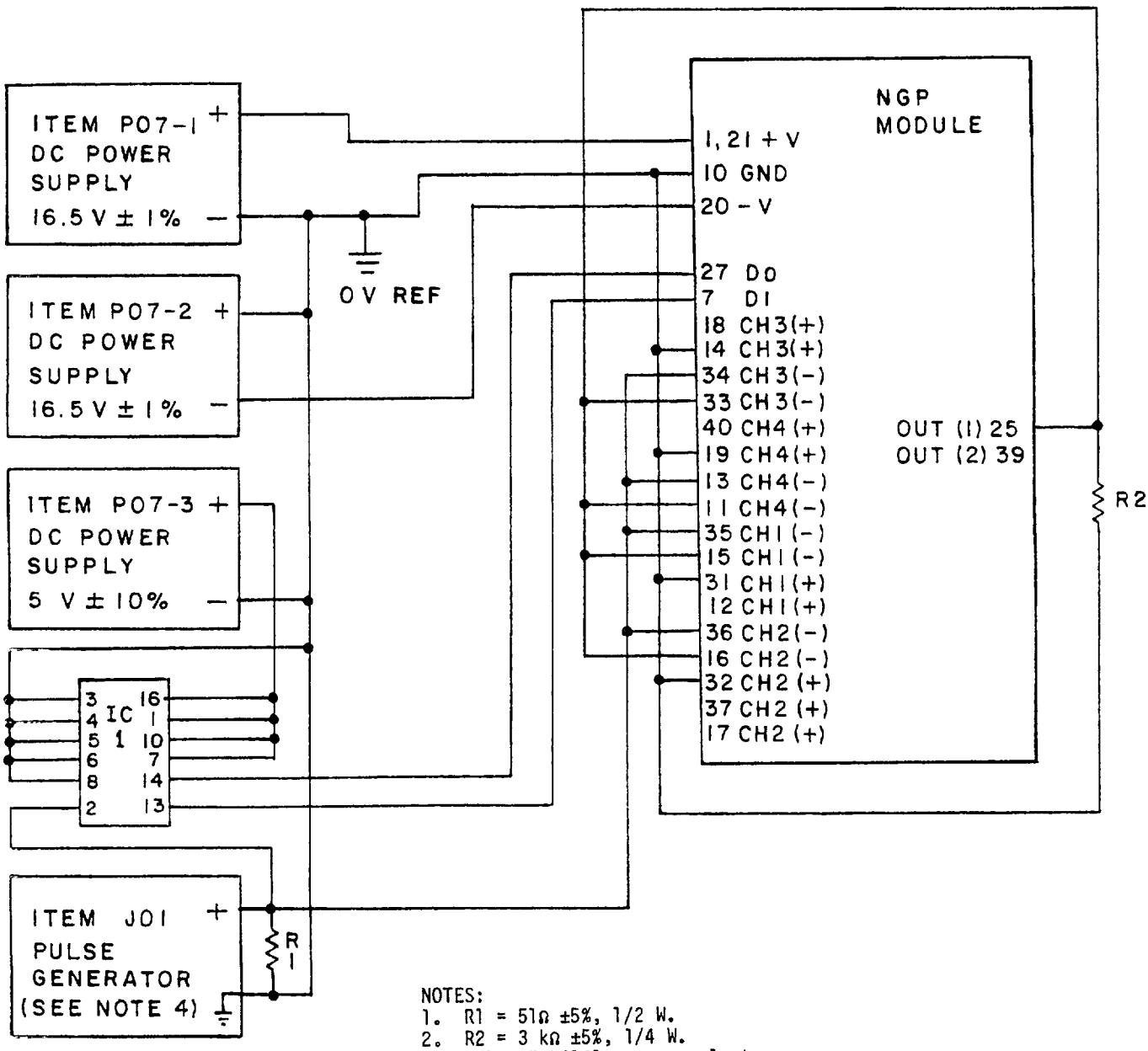
1/ L = Low level = 500 mV max.
2/ H = High level = 2.4 V min.

Agent:
DLA - ES

Preparing activity:
Navy - EC

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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 3 kΩ ±5%, 1/4 W.
3. IC1 = 74LS161 or equivalent.
4. Item JO1 setup:
   Frequency = 100 kHz ±10%.
   PA = 4.5 V ±1%.
   Duty cycle = 50 ±2%.
   $t_{TLH} = t_{THL} \leq 10$ ns.

FIGURE 2. Operating temperature, component temperature and life test circuit.

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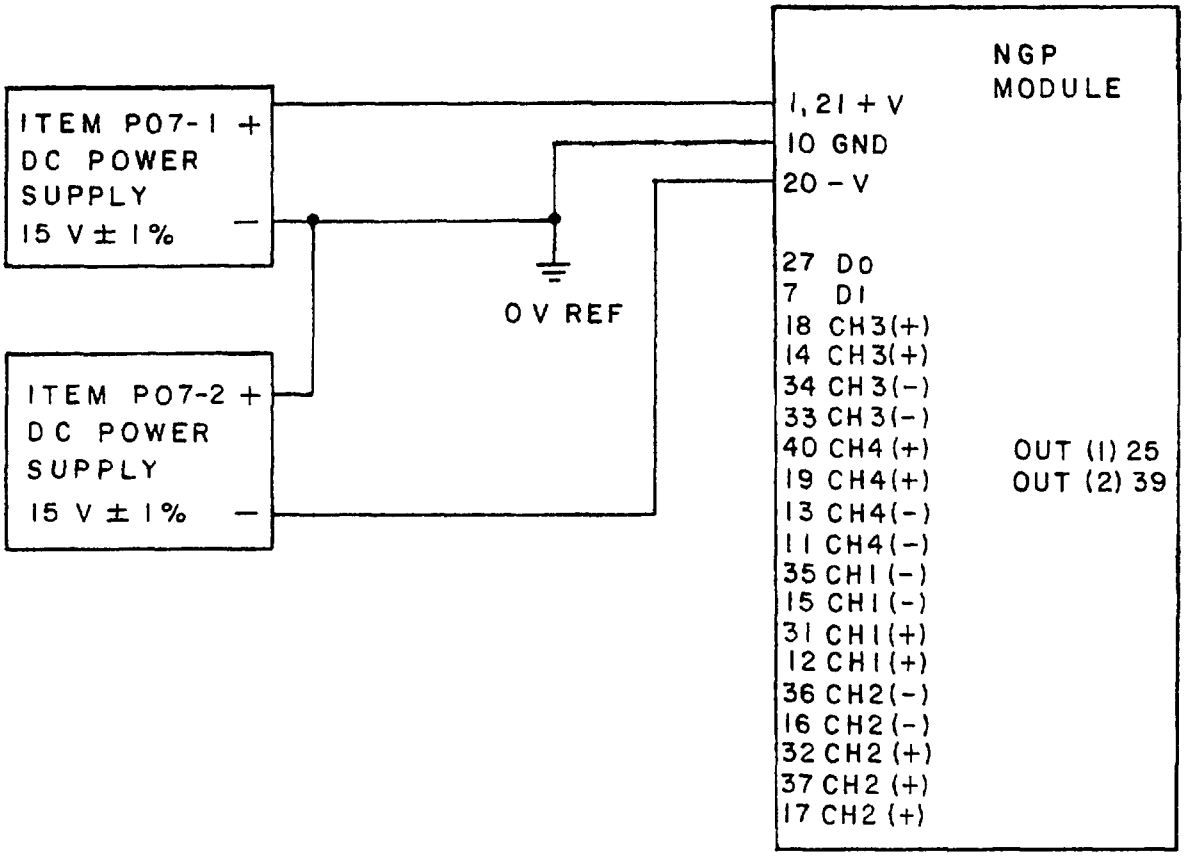


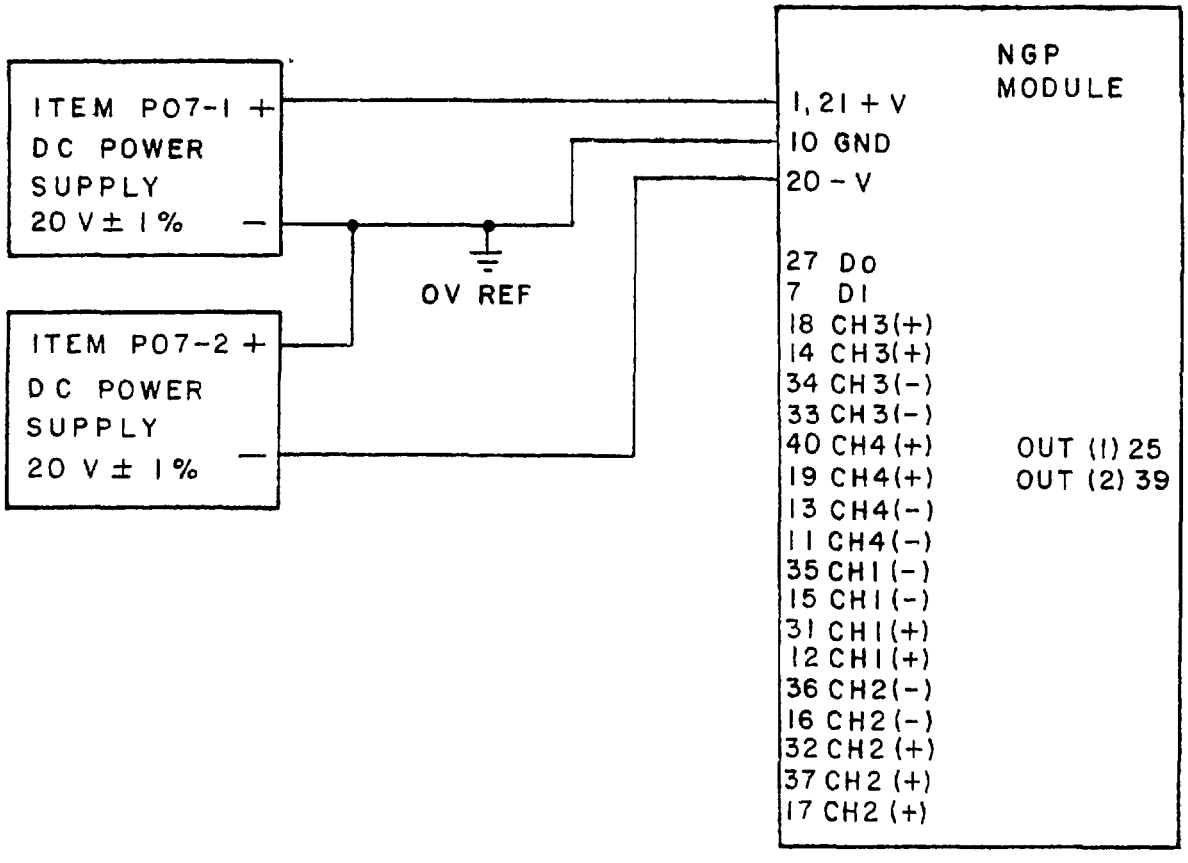
FIGURE 3. Vibration test circuit.

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TABLE III. V_S stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
A	V_S stress	Must pass remaining tests			4

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NOTE: Apply power for 1 minute minimum.

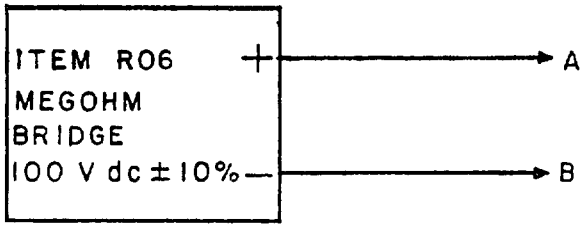
FIGURE 4. V_S stress test circuit for table III.

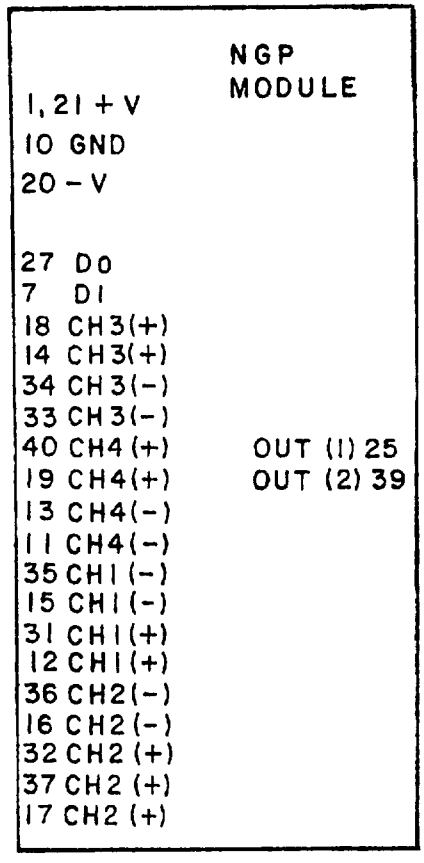
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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation	All pins	Frame	10	∞	MΩ	5
B2	"	1,3-40	2	"	"	"	"
B3	"	1,4-40	3	"	"	"	"
B4	"	1,5-40	4	"	"	"	"
B5	"	1,6-40	5	"	"	"	"
B6	"	1,7-40	6	"	"	"	"
B7	"	1,7,9-40	8	"	"	"	"
B8	"	1,7,10-40	9	"	"	"	"
B9	"	1,7,10-21,23-40	22	"	"	"	"
B10	"	1,7,10-21,24-40	23	"	"	"	"
B11	"	1,7,10-21,25-40	24	"	"	"	"
B12	"	1,7,10-21,25,27-40	26	"	"	"	"
B13	"	1,7,10-21,25-27,29-40	28	"	"	"	"
B14	"	1,7,10-21,25-26,30-40	29	"	"	"	"
B15	"	1,7,10-21,25-27,31-40	30	"	"	"	"
B16	"	1,7,10-21,25-27,31-37, 39,40	38	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition B1, penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table IV.

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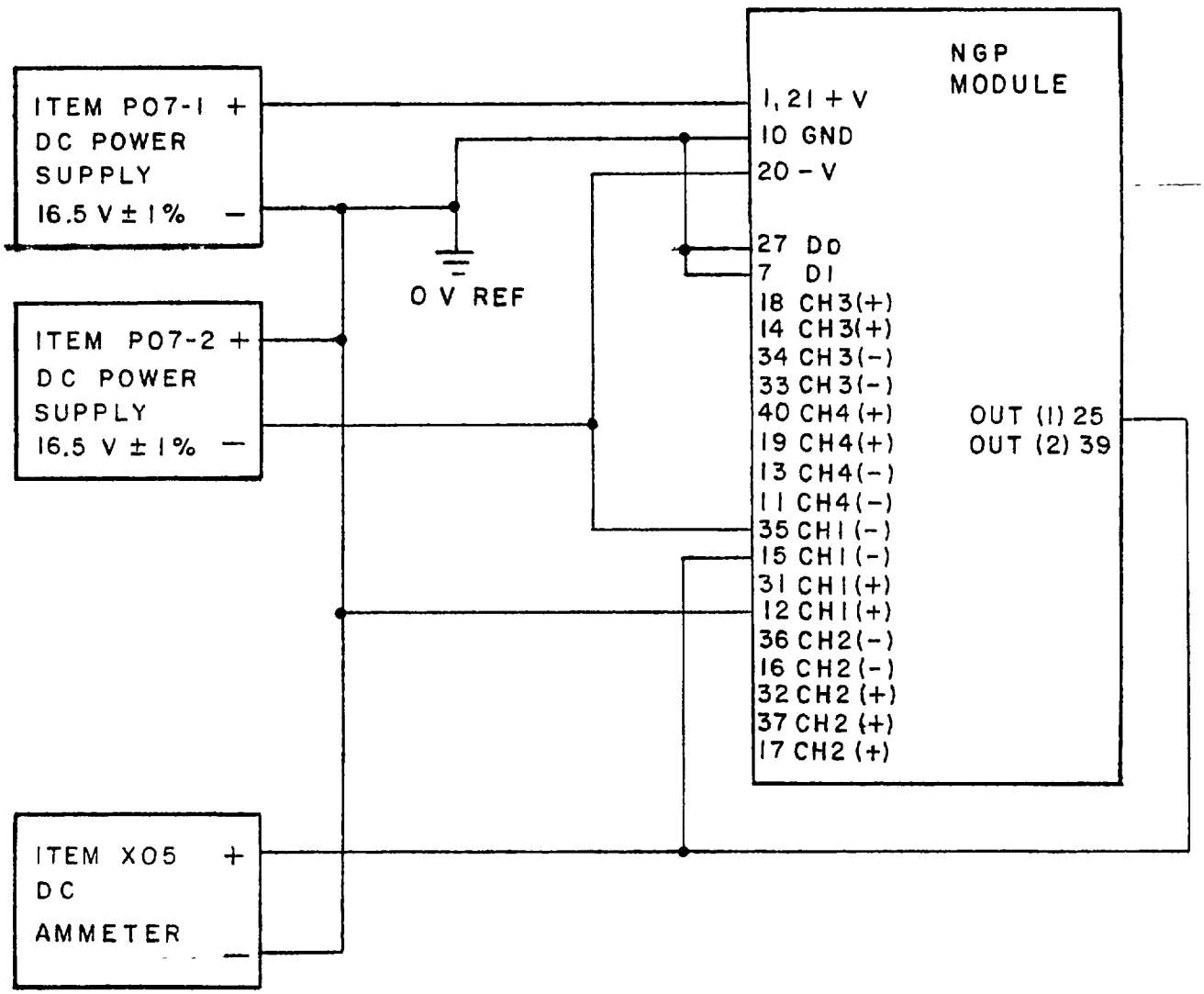
TABLE V. Output short circuit current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C	Output short circuit current	5	40	mA	6

TABLE VI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D1	Power supply current	1	8	mA	7
D2	Power supply current	1	8	mA	7

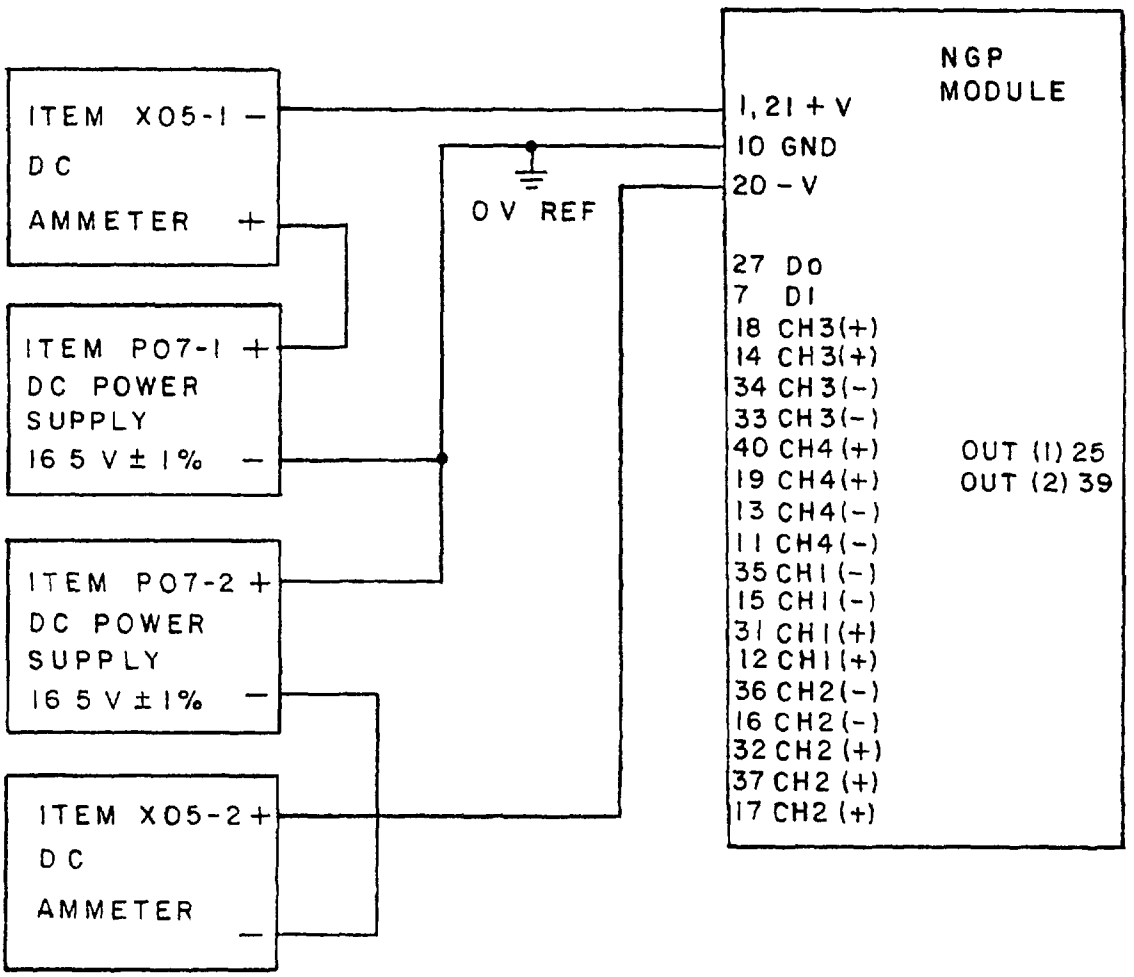
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NOTE: When performing output short circuit test, the measurement shall be taken after the power supplies have been applied for a minimum of 1 minute.

FIGURE 6. Output short circuit current test circuit for table V.

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NOTE:
1. Read item XO5-1 for test condition D1 and item XO5-2 for test condition D2.
2. Test conditions D1 and D2 are not run simultaneously; therefore, only one item XO5 is needed.

FIGURE 7. Power supply current test circuit for table VI.

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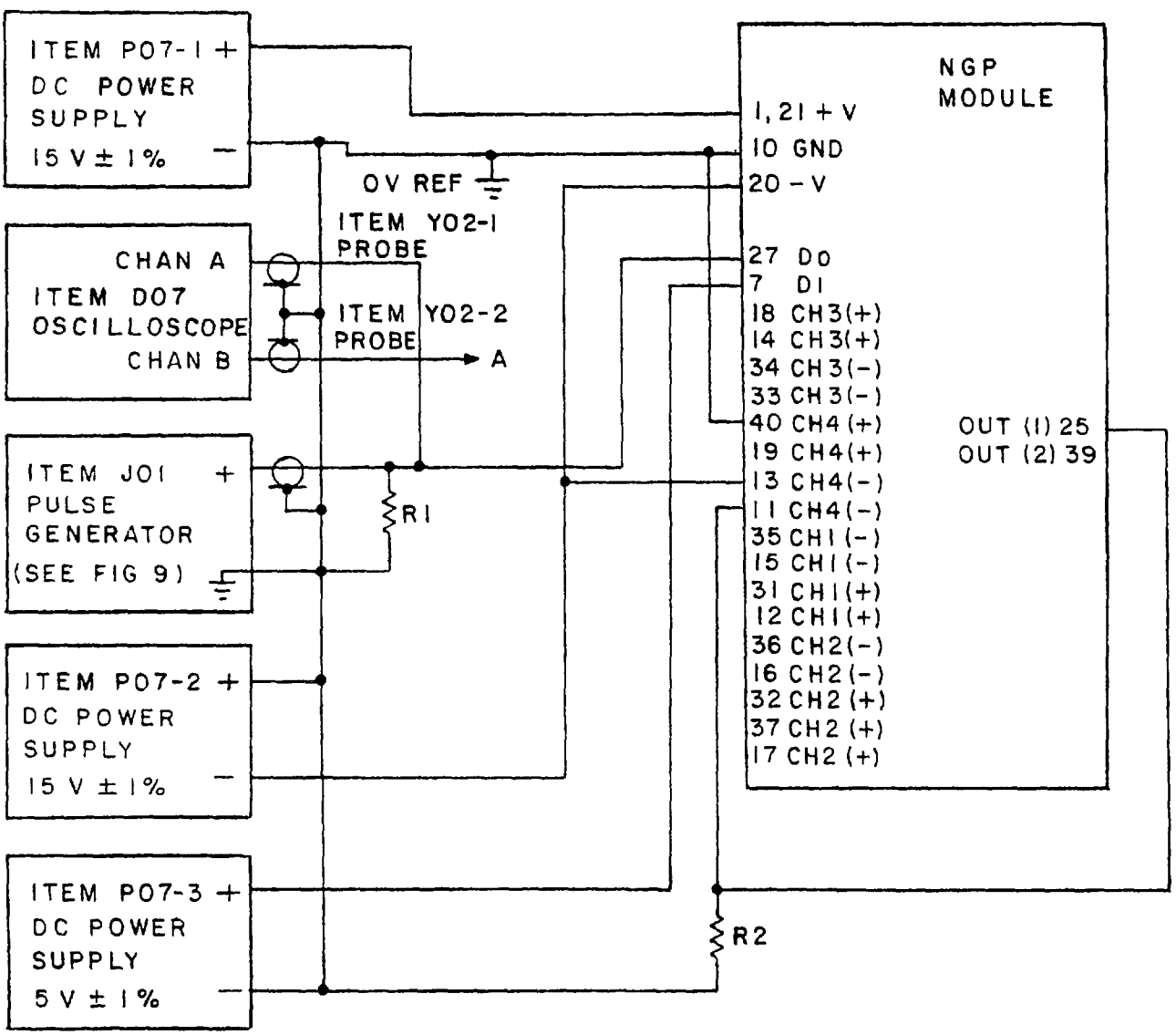
TABLE VII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	Power supply transient amplitude	21	0	350	mV	8,9
E2	Power supply transient amplitude	20	0	350	mV	8,9

TABLE VIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Power supply filter capacitance	1,21	10	0.75	1.25	µF	10
F2	Power supply filter capacitance	10	20	0.75	1.25	µF	10

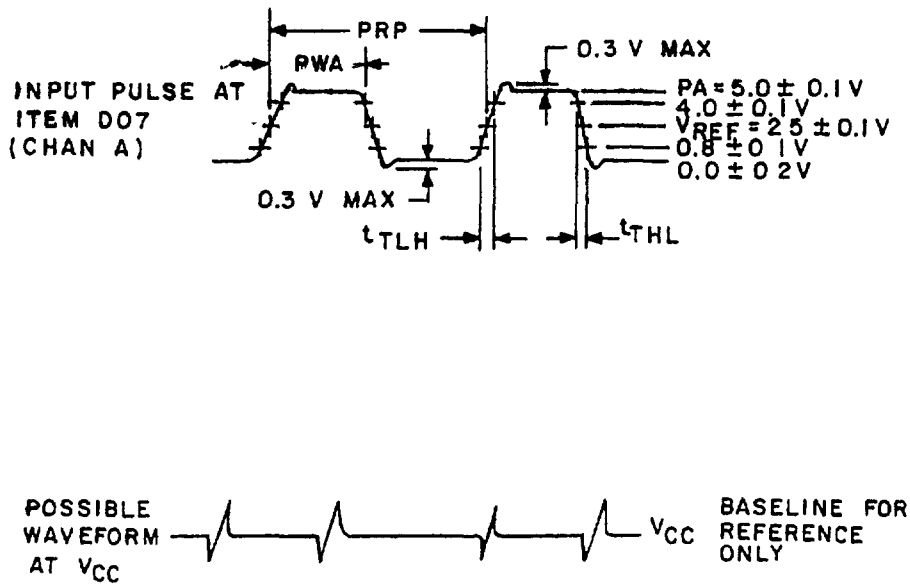
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 3 kΩ ±5%, 1/4 W.

FIGURE 8. Power supply transient amplitude test circuit for table VII.

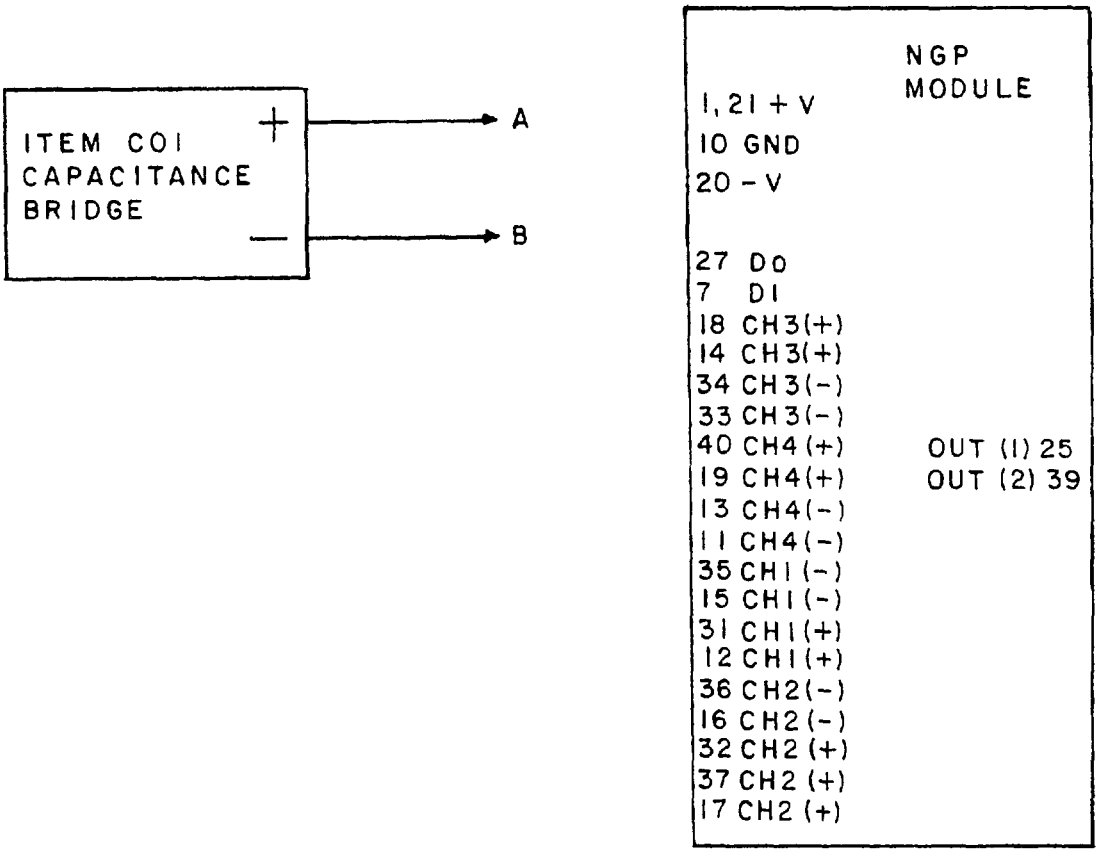
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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±1 ns (verified at each input pin).
2. For test condition E1 read most negative peak at pin 21 reference to baseline.
3. For test condition E2 read most positive peak at pin 20 reference to baseline.

FIGURE 9. Power supply transient amplitude test circuit waveforms for table VII.

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NOTE: Item C01 setup:
Frequency = 120 Hz ±10%.
V_{pk} = 0.3 V maximum.

FIGURE 10. Power supply filter capacitance test circuit for table VIII.

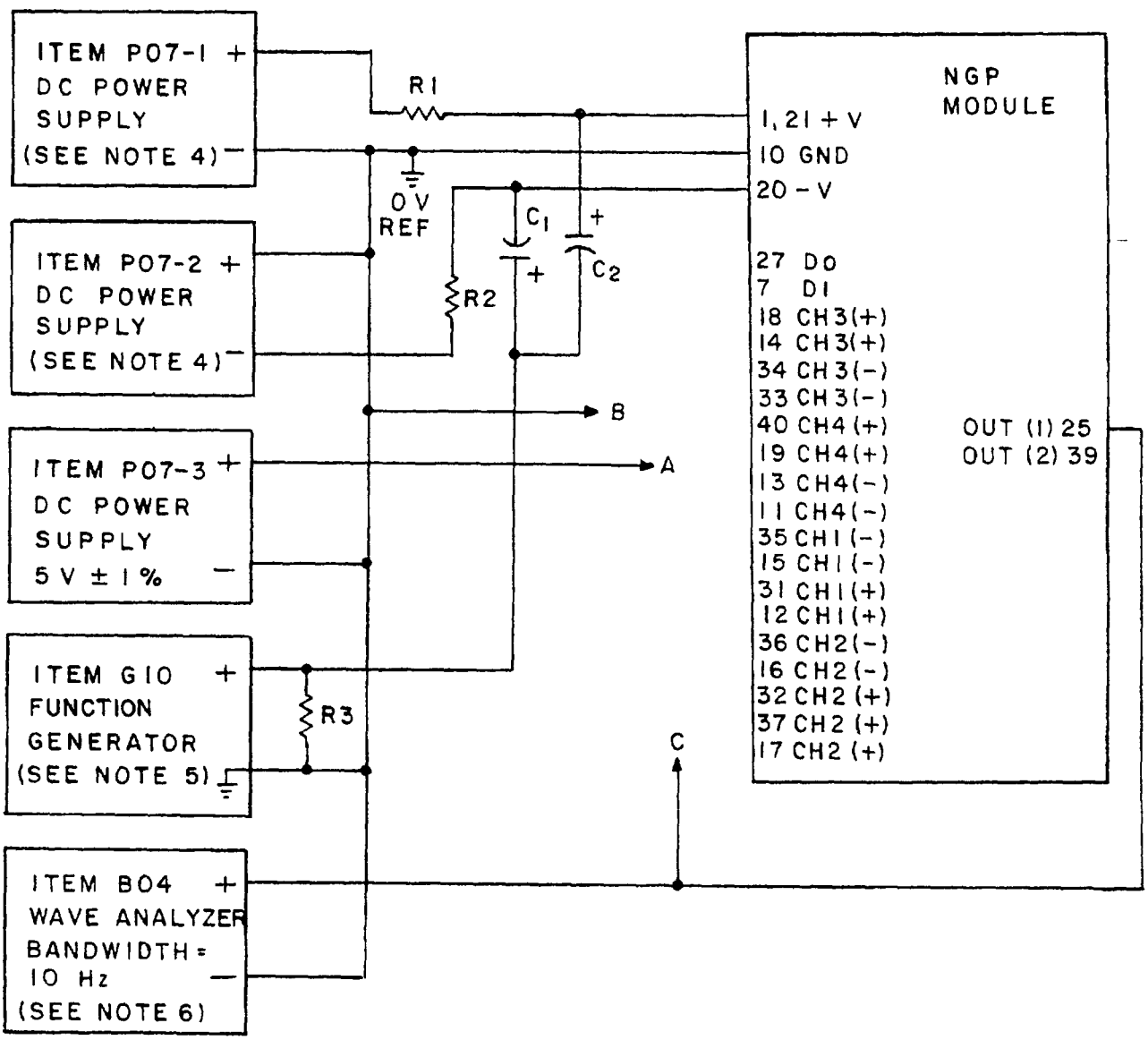
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TABLE IX. Power supply rejection ratio check test. 1/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Power supply rejection ratio check	Open	7,27,15,31,12	35	0.0	2.5	mV rms	11
G2	"	27	7,16,32,17,37	36	"	"	"	"
G3	"	7	27,14,34,18	33	"	"	"	"
G4	"	7,27	19,13,40	11	"	"	"	"

1/ PSRR (table I, dB) = 20 log $\left(\frac{V_{OUT}/Gain}{V_{IN}}\right)$ = 20 log $\left(\frac{\text{Item 804 reading}}{1\ V\ rms}\right)$.

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NOTES:
1. R1 = R2 = 100Ω ±5%, 1/4 W.
2. R3 = 51Ω ±5%, 1 W.
3. C1 = C2 = 20 µF ±20%, 20 V dc, electrolytic.
4. Adjust item P07-1 to obtain 15 V ±1% at pin 1; adjust item P07-2 to obtain 15 V ±1% at pin 20.
5. Adjust item G10 for 1 V rms ±5%, 1 kHz ±1% sine wave at pins 1 and 20.
6. Tune item B04 to 1000 Hz and fine tune for maximum deflection.

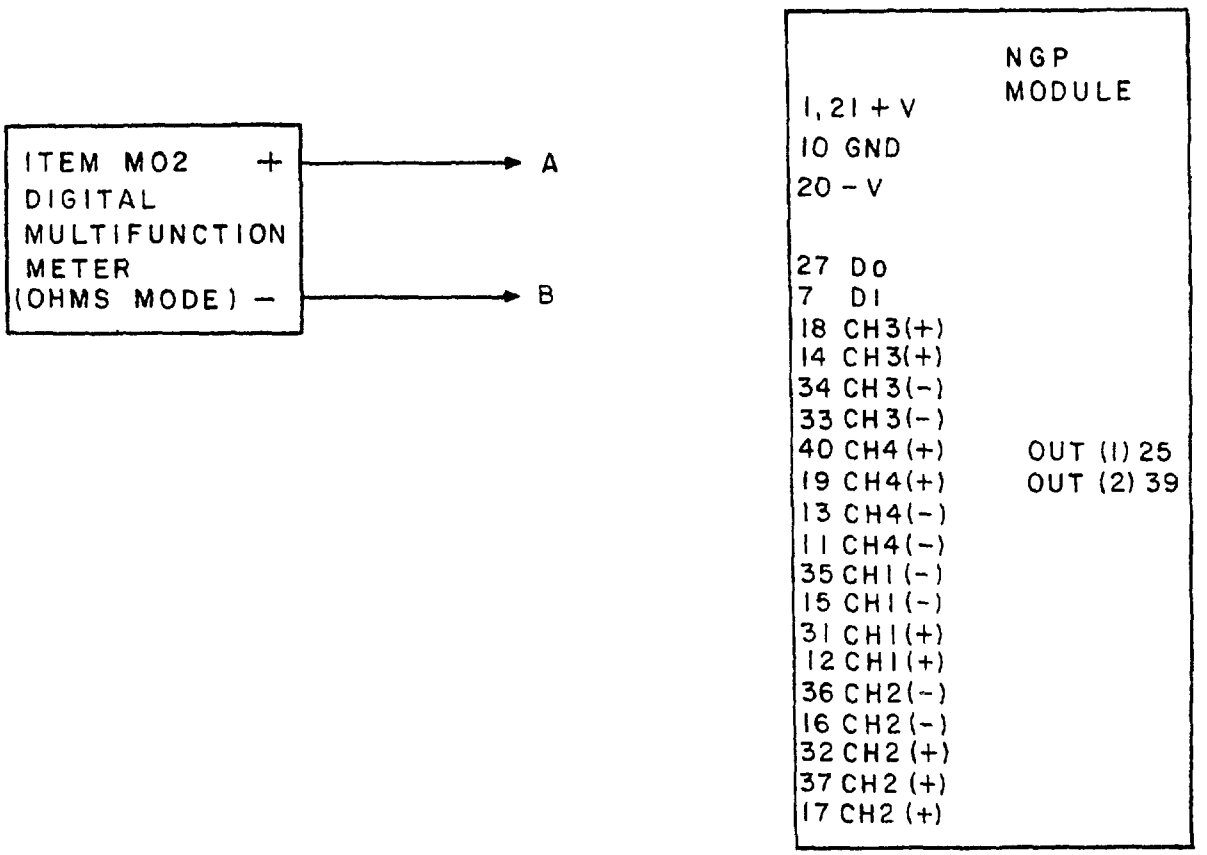
FIGURE 11. Power supply rejection ratio check test circuit for table IX.

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TABLE X. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	CH3(+) Resistance	18	14	101.4	103.0	kΩ	12
H2	CH3(-) "	34	33	"	"	"	"
H3	CH4(+) "	40	19	"	"	"	"
H4	CH4(-) "	13	11	"	"	"	"
H5	CH2(-) "	36	16	"	"	"	"
H6	CH2(+) "	32	37	"	"	"	"
H7	CH1(-) "	35	15	102.0	102.4	"	"
H8	CH1(+) "	31	12	102.0	102.4	"	"
H9	CH2(+) "	32	17	75.3	76.7	"	"
H10	OUT(1) "	25	10	9.0	9.1	"	"
H11	"	1	21	0	1	Ω	"
H12	OUT(2) "	39	10	2.99	3.03	kΩ	"

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NOTE: Use item M02 on 10 kΩ range for test condition H10.

FIGURE 12. Resistance test circuit for table X.

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TABLE XI. Offset voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Offset voltage	32,18,40	7,27	-18	18	mV	13
J2	"	27,12,18,40	7	"	"	"	"
J3	"	7,12,32,40	27	"	"	"	"
J4	"	7,27,12,32,18	Open	"	"	"	"

TABLE XII. Digital input current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Digital input current, high level	27	Open	0	15	nA	14
K2	Digital input current, high level	7	Open	0	15	nA	"
L1	Digital input current, low level	Open	27	0.1	1.6	mA	"
L2	Digital input current, low level	Open	7	0.1	1.6	mA	"

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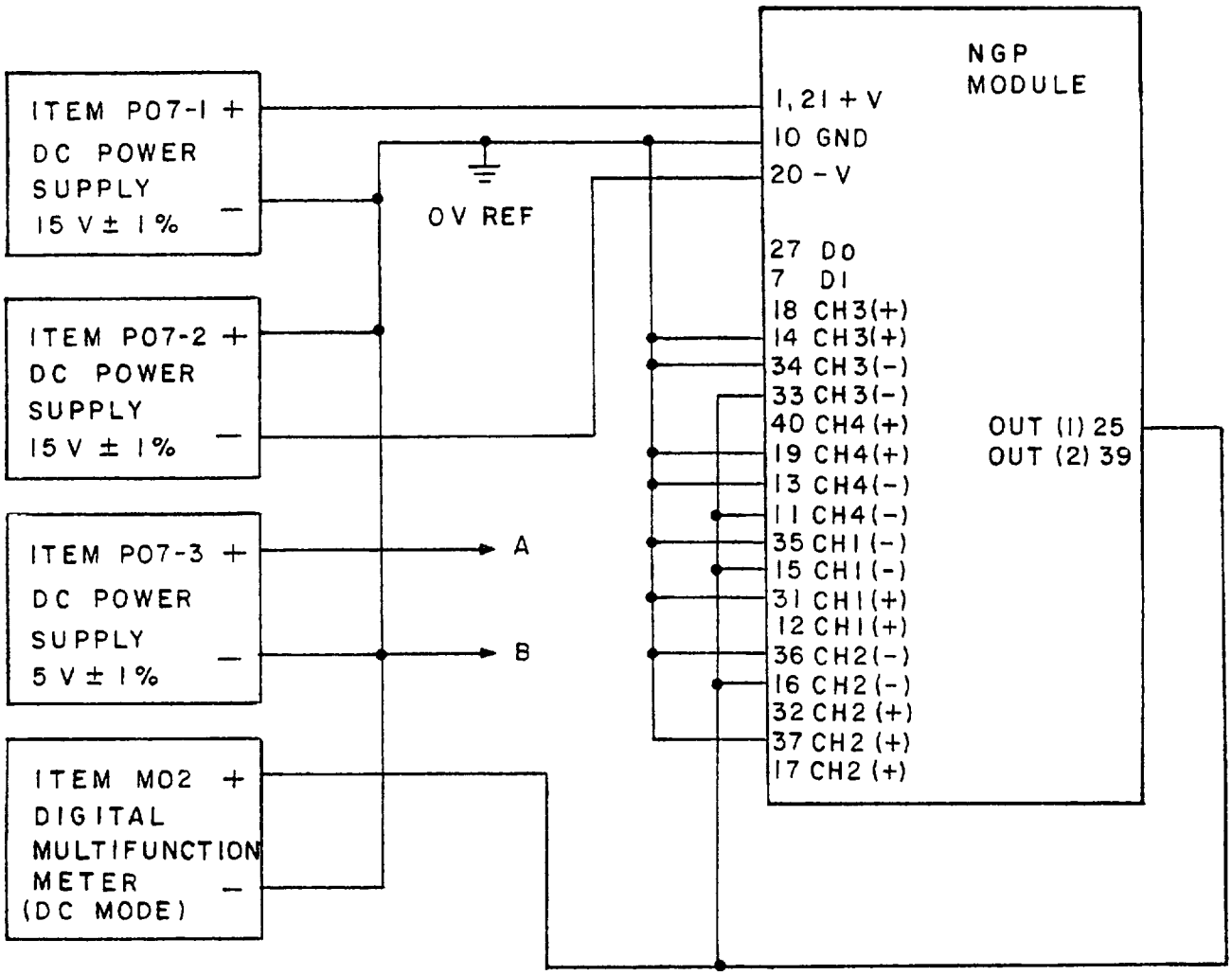
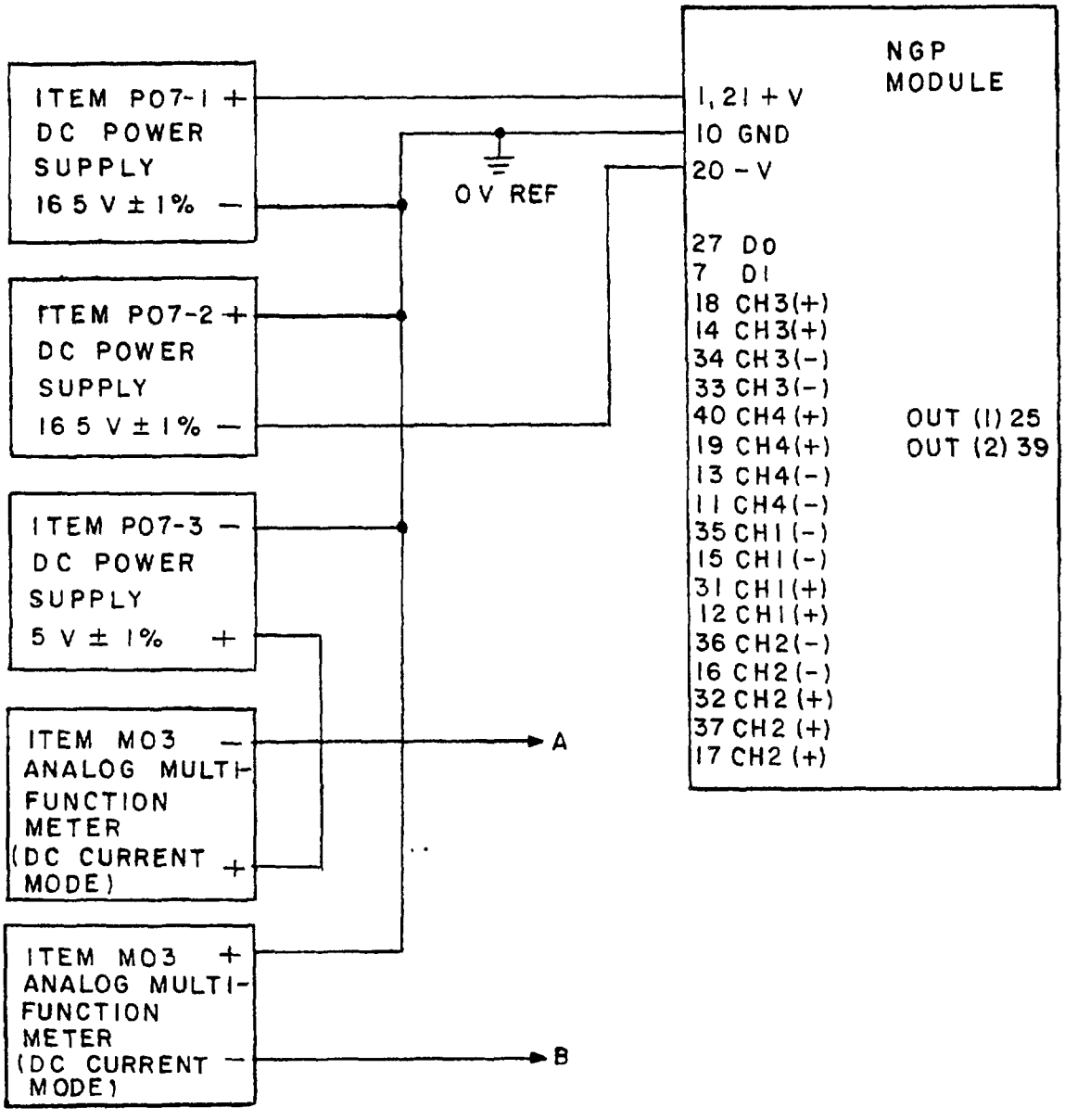


FIGURE 13. Offset voltage test circuit for table XI.

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NOTE: Since item M03-1 and M03-2 are not used simultaneously, only one item M03 is required.

FIGURE 14. Digital input current (high-level and low-level) test circuit for table XII.

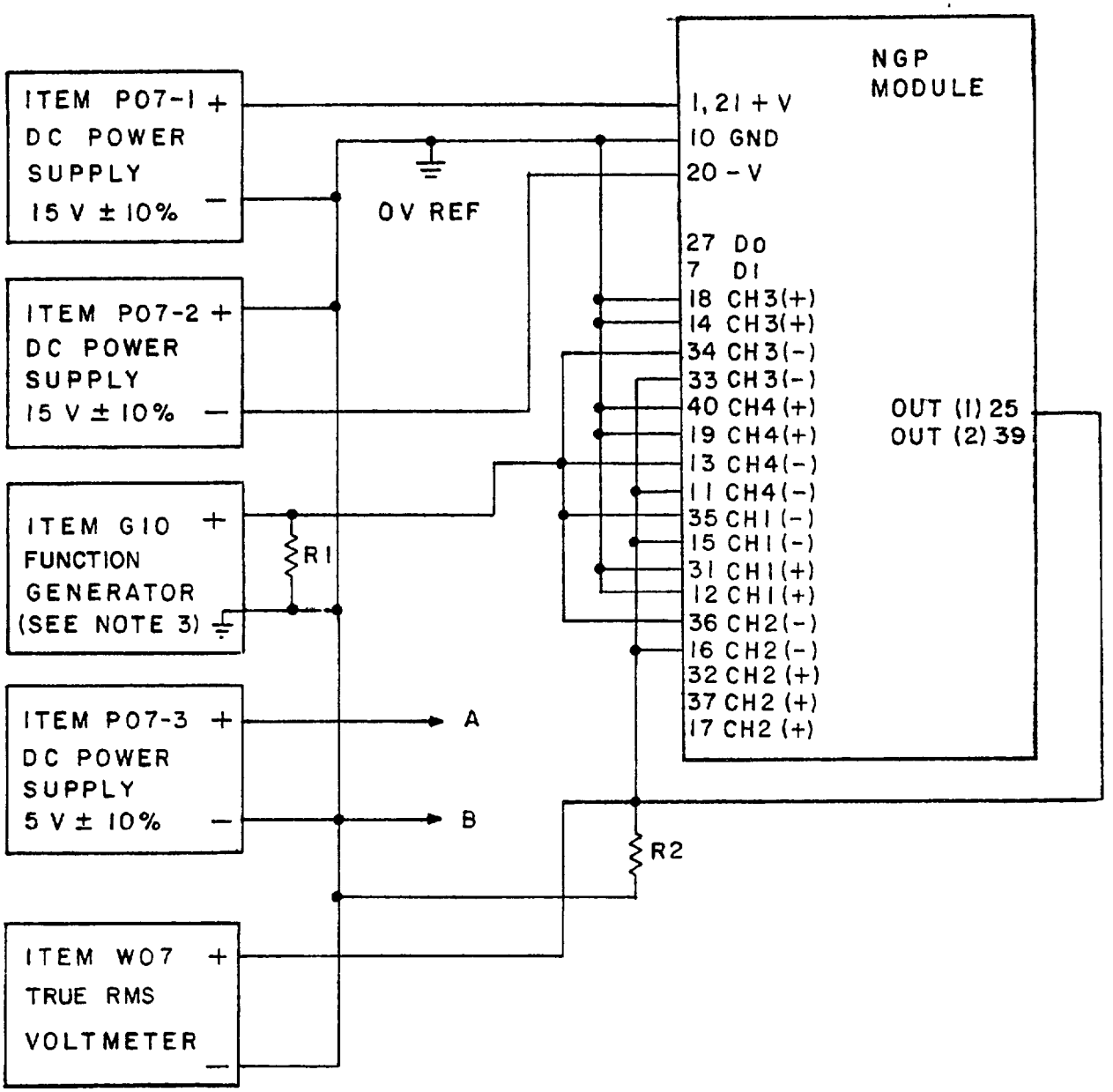
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TABLE XIII. Gain (200 kHz) check test. 1/ 2/

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Gain (200 kHz) check	Open	7,27,32,37	0.0707	0.1410	V rms	15
M2	"	27	7,32,37	"	"	"	"
M3	"	27	17,7	"	"	"	"
M4	"	7	27,32,37	"	"	"	"
M5	"	7,27	32,37	"	"	"	"

1/ This test verifies a minimum bandwidth of 200 kHz.
2/ Gain (table I, dB) = 20 log $\left(\frac{\text{Item M07 reading}}{0.1 \text{ V rms}}\right)$.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 3 kΩ ±5%, 1/4 W.
3. Adjust item G10 for a 0.1 V rms ±1%, 200 kHz ±1% sine wave.

FIGURE 15. Gain (200 kHz) check test circuit for table XIII.

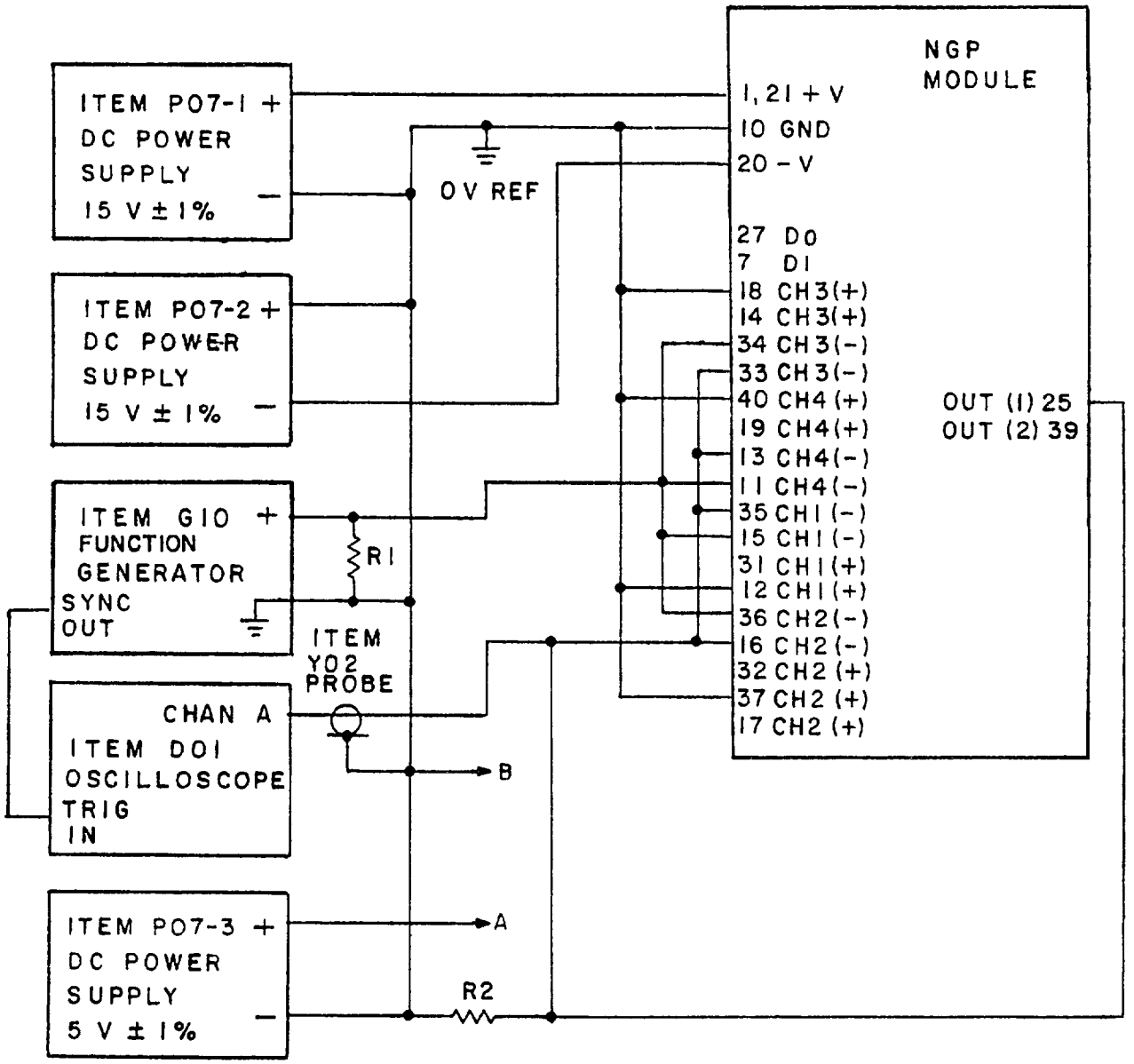
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TABLE XIV. Output slew rate check test. 1/

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Output slew rate check (negative)	Open	7,27	0.5	1.5	ns	16,17
N2	Output slew rate check (positive)	Open	7,27	"	"	"	"
N3	Output slew rate check (negative)	27	7	"	"	"	"
N4	Output slew rate check (positive)	27	7	"	"	"	"
N5	Output slew rate check (negative)	7	27	"	"	"	"
N6	Output slew rate check (positive)	7	27	"	"	"	"
N7	Output slew rate check (negative)	7,27	Open	"	"	"	"
N8	Output slew rate check (positive)	7,27	Open	"	"	"	"

1/ Slew rate (table I, V/μs) = $\frac{4\ V}{\text{Test condition N}}$.

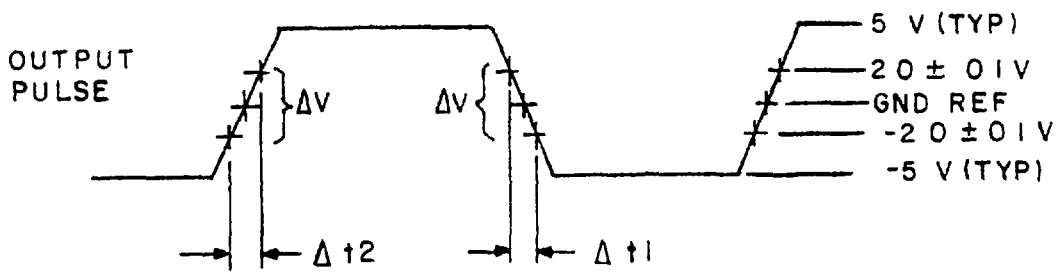
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 3 kΩ ±5%, 1/4 W.
3. Item G10 setup:
   PA = 10 V_{pp} ±1%.
   PRR = 1 kHz ±1%.
   $t_{TLH} = t_{THL} \leq 20$ ns.
   Square wave.

FIGURE 16. Output slew rate check test circuit for table XIV.

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NOTES:
1. ΔV/Δt = Output slew rate.
2. Δt1 = N1 - N7 (odd), negative slew rate.
3. Δt2 = N2 - N8 (even), positive slew rate.

FIGURE 17. Output slew rate check test circuit waveform for table XIV.

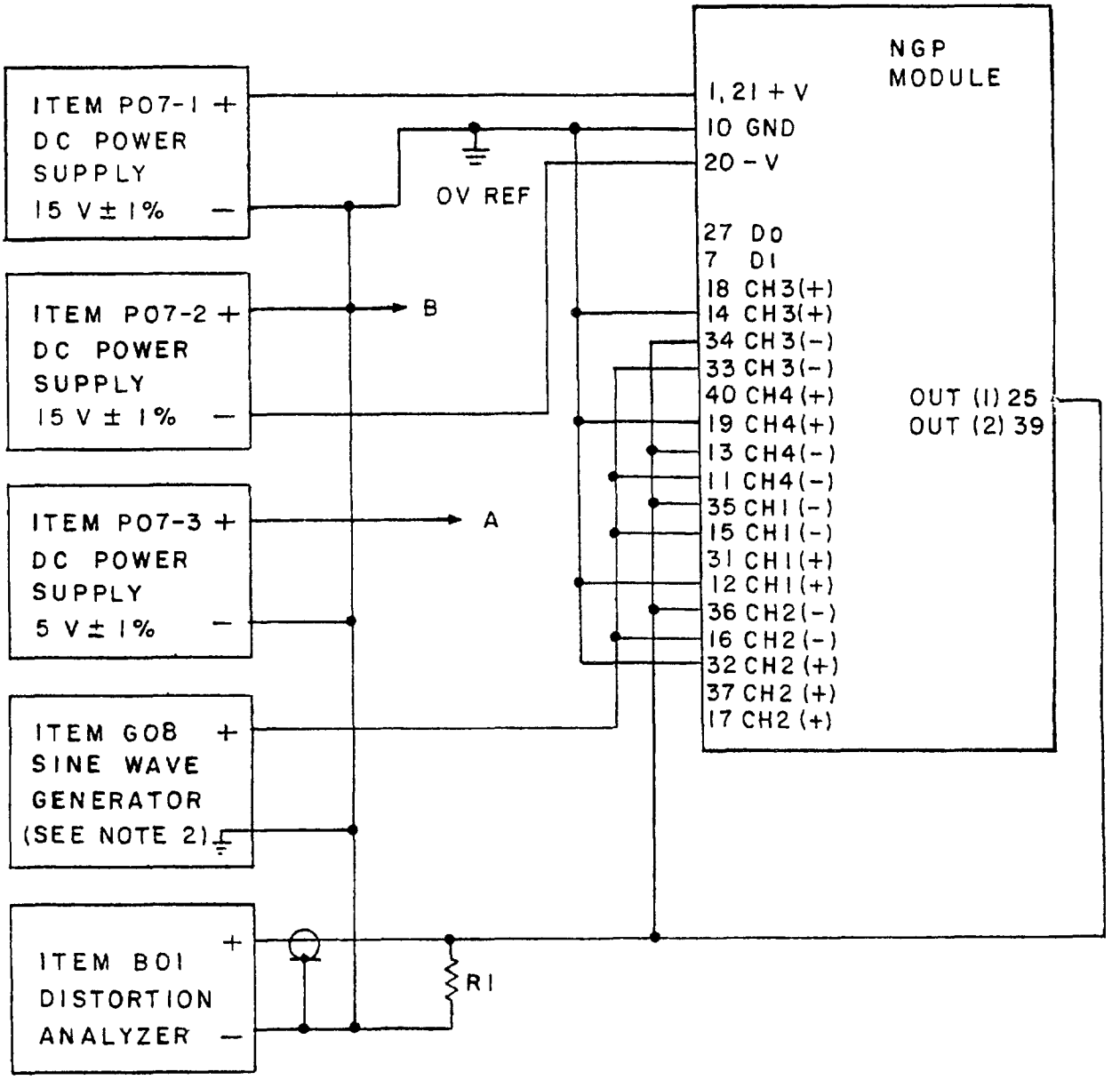
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TABLE XV. Distortion test. 1/

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Distortion	Open	7,27	0	1	%	18
P2	"	27	7	"	"	"	"
P3	"	7	27	"	"	"	"
P4	"	7,27	Open	"	"	"	"

1/ This test also verifies an output voltage swing of ±10 V.

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NOTES:
1. R1 = 3 kΩ ±5%, 1/4 W.
2. Item GO8 setup:
   Frequency = 1 kHz ±1%.
   Amplitude = 7.07 V rms ±1%.

FIGURE 18. Distortion test circuit for table XV.

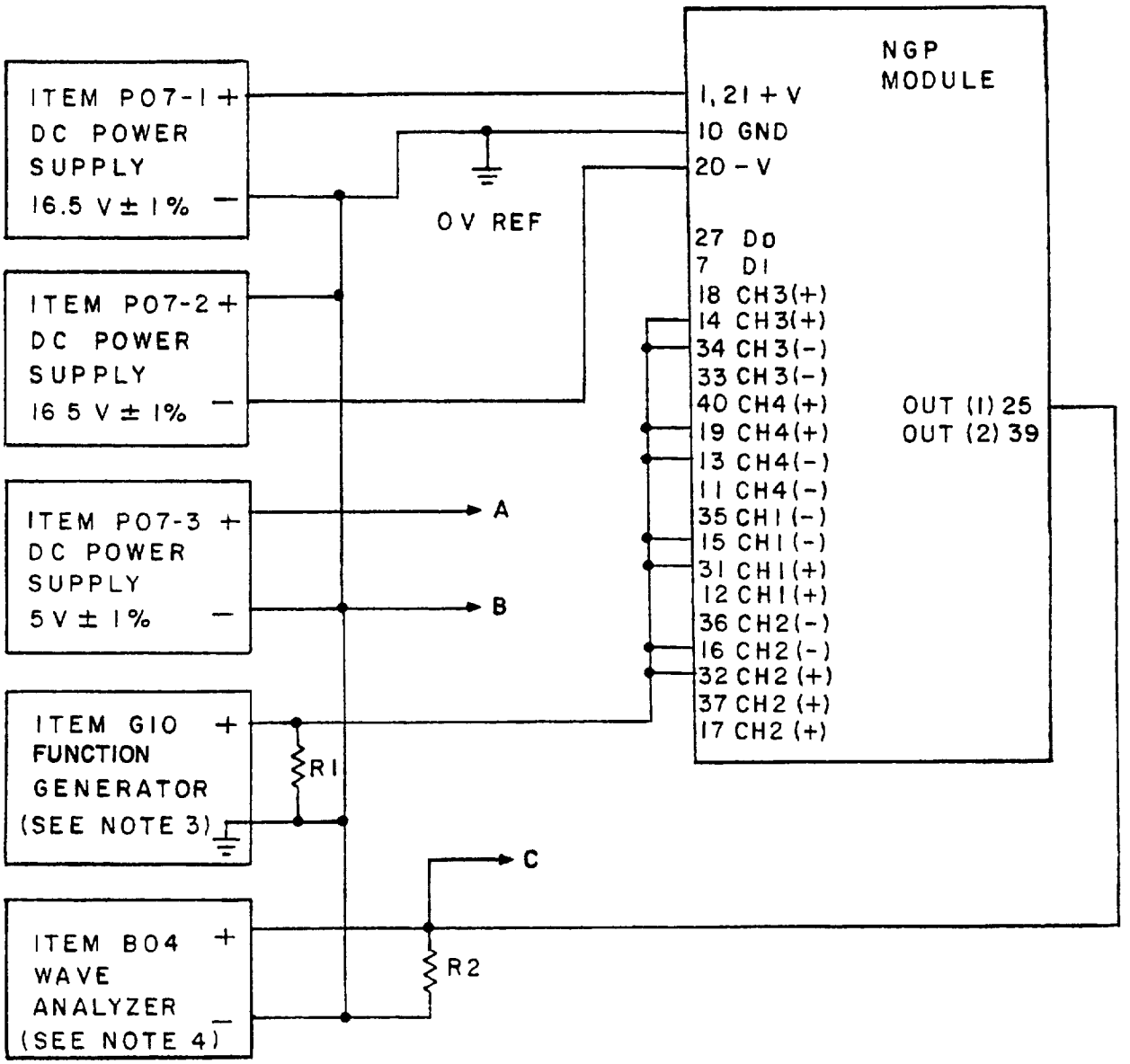
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TABLE XVI. Common mode rejection ratio check test. 1/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	Common mode rejection ratio check	Open	7,27,12	35	0.0	3.2	mV rms	19
R2	"	27	7,37	36	"	"	"	"
R3	"	7	27,18	33	"	"	"	"
R4	"	7,27	40	11	"	"	"	"

1/ CMRR (table I, dB) = $20 \log \left(\frac{V_{OUT}/Gain}{V_{IN}} \right) = 20 \log \left(\frac{\text{Item B04 reading}}{1 \text{ V rms}} \right)$.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 3 kΩ ±5%, 1/4 W.
3. Item G10 setup.
   Amplitude = 1 V rms ±1%.
   Frequency = 1 kHz ±1%.
   Sine wave.
4. Set item B04 to 10 Hz bandwidth. Tune to 1 kHz and then fine tune for a maximum deflection.

FIGURE 19. Common mode rejection ratio check test circuit for table XVI.

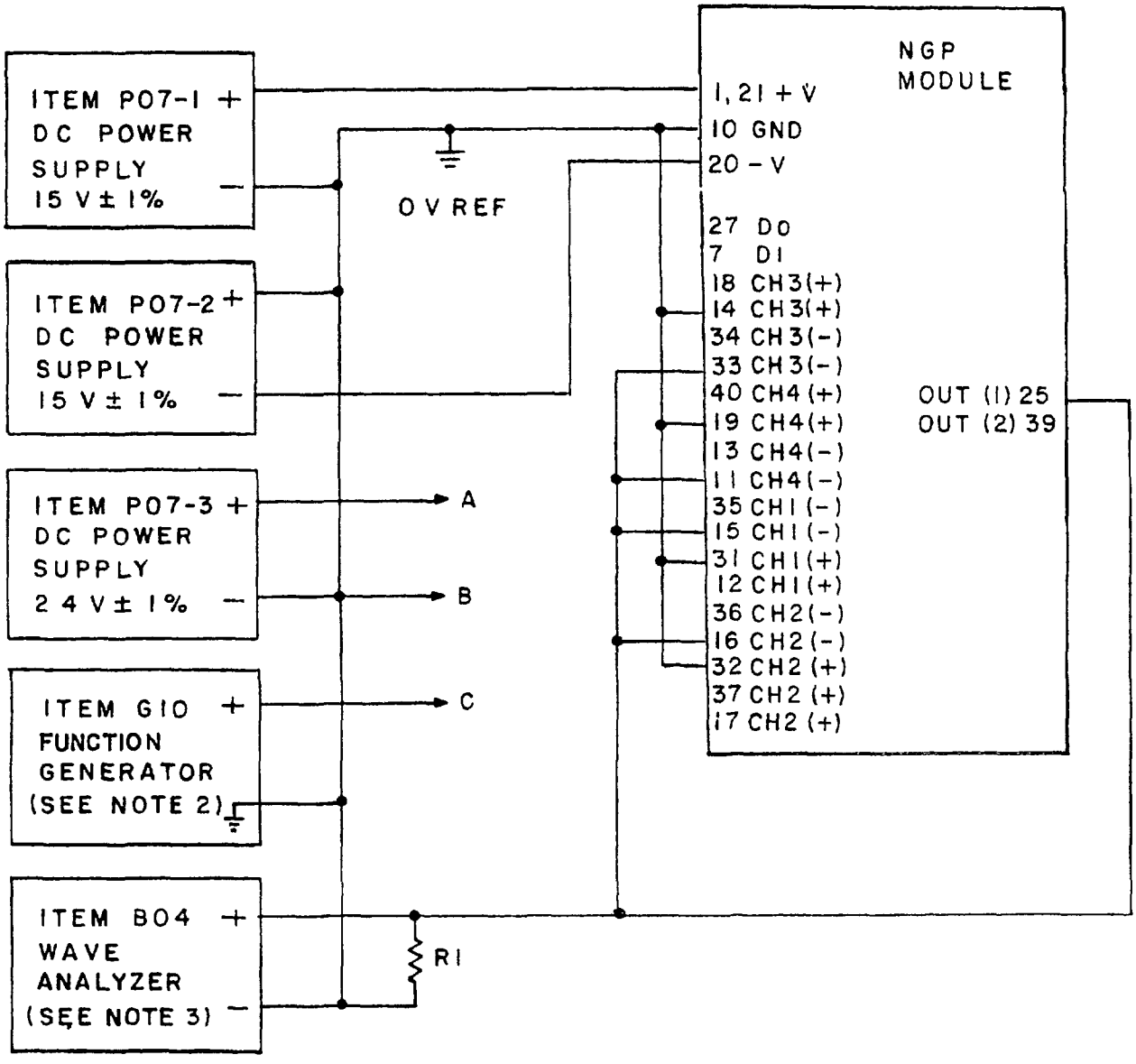
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TABLE XVII. Crosstalk check test. 1/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	Crosstalk check	Open	7,27	13,34,36	0	22	mV rms	20
S2	"	27	7	35,13,34	"	"	"	"
S3	"	7	27	13,35,36	"	"	"	"
S4	"	7,27	Open	34,35,36	"	"	"	"

1/ Crosstalk (table I, dB) = $20 \log \left(\frac{V_{OUT}/Gain}{V_{IN}} \right) = 20 \log \left(\frac{\text{Item B04 reading}}{7 \text{ V rms}} \right)$.

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NOTES:
1. R1 = 3 kΩ ±5%, 1/4 W.
2. Item G10 setup:
   Frequency 5 kHz ±1%.
   Amplitude 7 V rms ±1%.
   Sine wave.
3. Set item B04 to 10 Hz bandwidth. Tune to 1 kHz and then fine tune for a maximum deflection.

FIGURE 20. Crosstalk check test circuit for table XVII.

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TABLE XVIII. Functional test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	Functional	12	32,18,40	7,27	2.35	2.45	V	21
T2	"	27,32	12,18,40	7	"	"	"	"
T3	"	7,18	12,32,40	27	"	"	"	"
T4	"	7,27,40	12,32,18	Open	"	"	"	"

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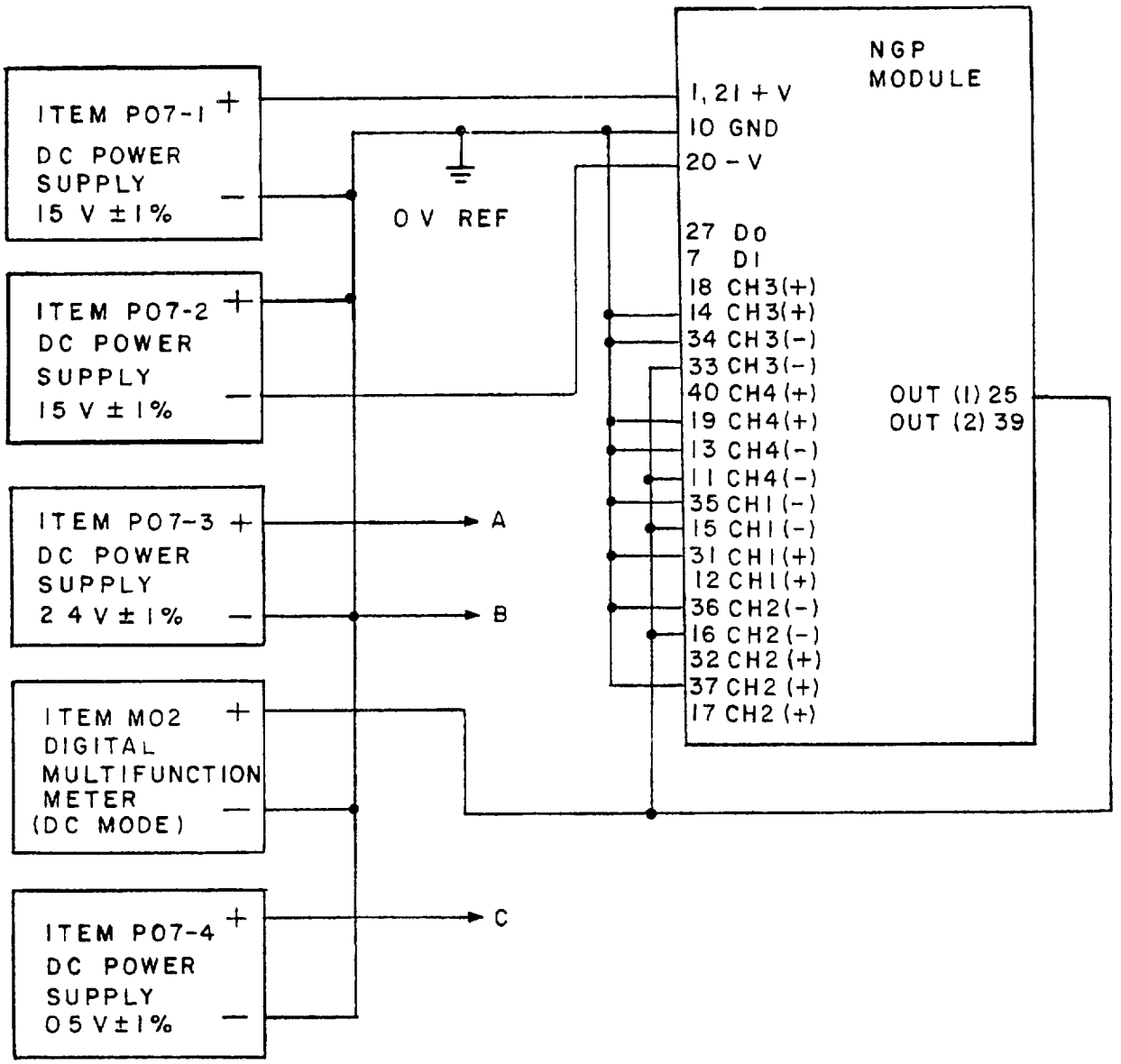


FIGURE 21. Functional test circuit for table XVIII.

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